

Development and evaluation of an Earth System Model with surface gravity waves

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[1] The critical role of oceanic surface waves in climate system is attracting more and more attention. We set up an Earth System Model, which is named as the First Institute of Oceanography-Earth System Model (FIO-ESM), composed of a coupled physical climate model and a coupled carbon cycle model. A surface wave model is introduced through including the nonbreaking wave-induced vertical mixing, which can improve the performance of climate model especially in the simulation of upper ocean mixed layer depth in the southern ocean, into the ocean general circulation model. The FIO-ESM is employed to conduct Coupled Model Intercomparison Project Phase 5 (CMIP5) experiments. The historical simulation of FIO-ESM's physical climate model for 1850–2005 shows that the simulated patterns of surface air temperature (SAT), rainfall, and El Niño-Southern Oscillation (ENSO) match those of the observations. Future projections under the four scenarios of RCP2.6, RCP4.5, RCP6.0, and RCP8.5 are also conducted and the global averaged SAT in 2100 would be -0.007°C , 1.10°C , 1.85°C , and 3.92°C higher than that in 2005, respectively. The historical simulation and future projection under RCP8.5 with global carbon cycle show the SAT and atmospheric CO_2 concentration are well reproduced in the historical period and the global averaged SAT would increase by 3.90°C in 2100, which is quite similar to the physical climate model's result. Further analysis shows surface wave makes projected SAT in RCP2.6 about 2°C cooler in the Arctic area and 2°C warmer in the southern ocean.

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1. Introduction

[2] State-of-the-art climate models normally contain four components of atmospheric general circulation model (AGCM), ocean general circulation model (OGCM), land surface model, and sea ice model, which interact with each other. Since the pioneer work on establishing a climate model by *Manabe and Bryan* [1969], climate models have archived great progress, especially through several important climate model intercomparison programs such as the Coupled Model Intercomparison Project (CMIP), the Paleoclimate Modelling Intercomparison Project (PMIP), and the Cloud Feedback Model Intercomparison Project (CFMIP). Climate models have become more and more important for climate process study and projection of the effect of anthropogenic activities on the earth climate. The main stream of climate model development focuses on

higher resolution, more accurate parameterizations of unresolved physical processes, and including more biogeochemical processes.

[3] Although as early as 1991, *Hasselmann* [1991] noted that surface gravity waves should be included in the Earth System Models (ESMs), most researchers regard the spatial and temporal scales of surface waves being too small to be included in a climate model. However, surface waves play key roles in the following two crucial processes. First, the breaking waves influence the air-sea fluxes. The drag coefficient is closely related to surface waves [*Hasselmann*, 1991; *Babanin*, 2011]. By including ocean surface waves in an AGCM, *Yu et al.* [2005] improved the simulation of global sea level pressure. At the same time, breaking waves can affect the air-sea heat flux and gas transfer, such as water vapor, CO_2 , and aerosol fluxes [*Wunsch and Ferrari*, 2004; *Cavaleri et al.*, 2012]. Through efforts on parameterization of air-sea fluxes, the surface waves can much reduce the uncertainty of air-sea flux estimations [*Guan and Xie*, 2004], which implies that surface waves are important factor in parameterizing the air-sea fluxes. Second, surface waves play a dominant role in the upper ocean through vertical mixing induced by both breaking and nonbreaking waves. As the first step, the breaking wave-induced vertical mixing is considered [*Craig and Banner*, 1994; *Mellor and Blumberg*, 2004]; this type of mixing can only affect a few

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meters in the upper ocean, which is the depth in the order of wave amplitude. The nonbreaking wave-induced vertical mixing (Bv), confirmed by laboratory experiments [Babanin and Haus, 2009; Dai et al., 2010; Savelyev et al., 2012; Toffoli et al., 2012], plays a key role in the upper ocean [Qiao et al., 2004, 2010], and can effectively improve different kinds of ocean models in the simulation of the upper ocean temperature [Huang and Qiao, 2012]. It is interesting that even excluding the shear-induced mixing in the Mellor-Yamada scheme [Mellor and Yamada, 1982] Bv can work well in a global OGCM [Qiao and Huang, 2012]. As the ocean is a key component of the climate system, it is natural to improve climate models by including surface wave component in the climate system.

[4] As case studies, Song et al. [2007] and Huang et al. [2008] included precalculated Bv, i.e., to run a surface wave model separately for obtaining Bv, the sea surface temperature (SST) and temperature structure in the upper ocean of the flexible coupled general circulation model [Yu et al., 2002] was dramatically improved, for example, the global averaged SST deviation of 1.2°C was cured. De Szoeke and Xie [2008] compared the results of 15 coupled general circulation models (CGCMs) submitted to the CMIP3 and found that most of these models simulated two cold phases in the eastern equatorial Pacific SST seasonal cycle rather than a single one as observed. Following previous work and as a test, Song et al. [2011] put the marine science and numerical modeling (MASNUM) surface wave model into the Community Climate System Model Version 3 (CCSM3), which is one of the state-of-the-art climate models in the CMIP3. Their numerical experiments showed that surface waves can successfully remove the spurious semiannual cycle of the eastern equatorial Pacific SST simulated by the original CCSM3. Another common problem of climate models without flux correction is that the simulated SST deviates noticeably from the observation in the tropical areas, including the cold tongue in the eastern Pacific being too cold and a reversed SST zonal gradient in the equatorial Atlantic. By incorporating online Bv into the CCSM3 through a coupler, the simulated tropical SST was much improved [Song et al., 2012b]. The mixed layer depth (MLD) in the upper ocean is one of the key factors controlling air-sea interaction and biogeochemical processes, and accurate simulation of summer MLD remains a challenge for most climate models, especially in the Southern Ocean. Huang et al. [2012] showed that Bv can dramatically improve the performance of ocean circulation models and climate model CCSM3 in terms of MLD simulation. With a similar idea but different scheme, Babanin et al. [2009] parameterized the MLD due to wave-induced turbulence and Langmuir circulation triggered by the surface waves in terms of global winds; and their climate model results showed that the seasonal temperature modulations and extremes were significantly improved. Belcher et al. [2012] noted that the vertical mixing in the upper ocean is not strong enough in climate models, and that surface wave-related vertical mixing should be introduced into the climate models. Yuan and Huang [2012] thoroughly reviewed the research of surface waves, and emphasized the critical importance of including surface waves in large-scale geophysical fluid dynamics. All the above mentioned studies suggest that including surface wave-induced vertical mix-

ing through a coupler should be an effective way to improve the performance of climate models. Our effort presented in this paper should be the first to include surface waves in a climate model among all the climate models participated in the CMIP5.

[5] The rest of the paper is structured as follows. Section 2 presents a new climate model. Model results and validations are given in section 3, and the summary is in section 4.

2. Earth System Model Description and Numerical Experiments Design

2.1. Earth System Model Development

[6] The FIO-ESM Version 1.0 (FIO-ESM V1.0) contains two parts: coupled physical climate model and carbon cycle model (Figure 1). The coupled physical climate model includes the atmosphere, land surface, sea ice, ocean, and surface wave model components, which are organized through coupler 6 developed by the National Center for Atmospheric Research (NCAR). The components are the Community Atmosphere Model Version 3 (CAM3) [Collins et al., 2006], the Community Land Model Version 3.5 (CLM3.5) [Dickinson et al., 2006; Oleson et al., 2008], the Los Alamos National Laboratory sea ice model Version 4 (CICE4) [Hunke and Lipscomb, 2008], the Parallel Ocean Program Version 2.0 (POP2.0) [Smith et al., 2010], and the MASNUM surface wave model [Yang et al., 2005]. The horizontal resolutions of the CAM3 (with 26 vertical layers) and CLM3.5 are T42 spectral truncation (about 2.875°), a nominal 1° (about 1.1° in longitude and 0.3–0.5° in latitude, with the North Pole displaced to the Greenland) for POP2.0 (with 40 vertical layers) and CICE4, and 2° by 2° for the MASNUM surface wave model (with a resolution of 30° for wave direction). The global carbon cycle model coupled with the physical climate includes land carbon cycle model Carnegie-Ames-Stanford approach (CASA) [Potter et al., 1993; Doney et al., 2006], ocean carbon cycle model modified OCMIP-2 [Najjar and Orr, 1998; Bao et al., 2012], and three-dimensional transport of CO₂ in the atmosphere. The atmospheric CO₂ concentration is controlled by the air-sea and air-land CO₂ fluxes, and also the anthropogenic emissions through equation (1).

$$\frac{\partial}{\partial t} CO_2 + L(CO_2) = F_{AO} + F_{AL} + J_{CO_2}, \quad (1)$$

where $L(CO_2)$ represents the physical processes, F_{AO} and F_{AL} are the air-sea CO₂ flux and air-land CO₂ flux, respectively, J_{CO_2} is the anthropogenic CO₂ emissions including the fossil fuel burning emission and land use change emission after the industrial revolution. The air-sea CO₂ flux F_{AO} and air-land CO₂ flux F_{AL} are calculated in the ocean and land component models, respectively, and are then sent to the atmosphere component model through the coupler.

[7] Atmosphere, land surface, and sea ice components exchange data with the coupler each hour, while ocean and surface wave components exchange data with coupler each 24 and 6 h, respectively. For the MASNUM surface wave model, it gets surface wind at 10 m height from coupler each 6 h, integrates to produce wave number spectrum,

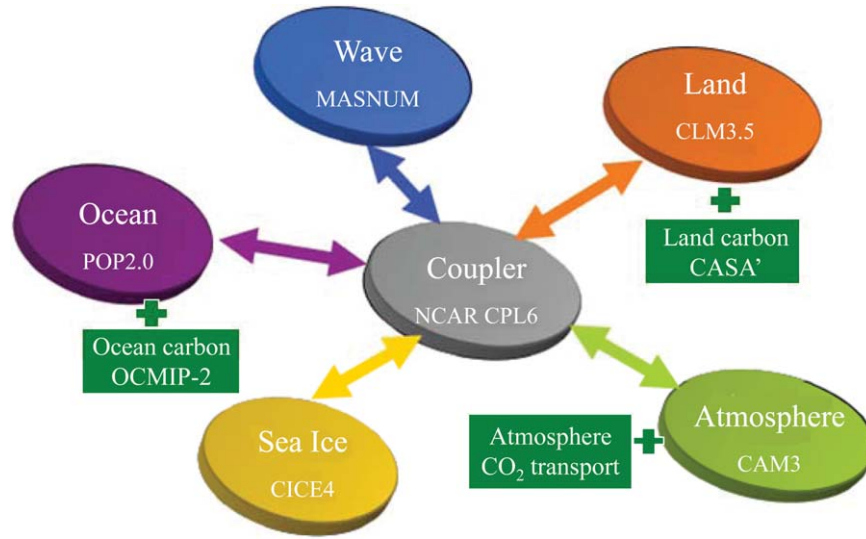


Figure 1. A schematic view of the First Institute of Oceanography's-Earth System Model (FIO-ESM, Version 1.0).

calculates nonbreaking wave-induced vertical mixing (B_v) through equation (2) below [Qiao *et al.*, 2004], and finally sends B_v back to the coupler. The ocean model gets daily averaged B_v from the coupler each 24 h, and adds it to the original vertical viscosity and diffusivity through equations (3a) and (3b).

$$B_v = \alpha \int \int_{\vec{k}} E(\vec{k}) \exp\{2kz\} d\vec{k} \frac{\partial}{\partial z} \left(\int \int_{\vec{k}} \omega^2 E(\vec{k}) \exp\{2kz\} d\vec{k} \right)^{1/2} \quad (2)$$

where α is a constant coefficient usually set to 1.0, $E(\vec{k})$ represents the wave number spectrum, ω is the wave angular frequency, $\vec{k}$ is the wave number, and z is the vertical coordinate (upward being positive) with $z = 0$ at the ocean surface.

[8] The vertical viscosity K_m and diffusivity K_h are,

$$K_m = K_{m0} + B_v \quad (3a)$$

$$K_h = K_{h0} + B_v \quad (3b)$$

where K_{m0} and K_{h0} are viscosity and diffusivity from the original OGCM of POP2.0. In our study, they are calculated from the K profile parameterization (KPP) scheme [Large *et al.*, 1994].

2.2. Numerical Experiments Design

[9] The present work adopts the numerical experiments design and forcing data suggested by the CMIP5 [Taylor *et al.*, 2012]. Representative Concentration Pathways (RCPs) are four greenhouse gas trajectories adopted by CMIP5. The four RCPs, RCP2.6, RCP4.5, RCP6.0, and RCP8.5, are named after a possible range of radiative forcing values in the year 2100 (2.6, 4.5, 6.0, and 8.5 W/m^2 , respectively). The control run for the preindustrial period

(before 1850 AD) integrates the coupled physical climate model for 1200 years with the constant forcing fields of greenhouse gases, aerosol, and solar irradiance from 1850. The evolution of global mean SST, a key variable of the climate system, indicates that the climate system adjusts rapidly within 40 years (Figure 2). After having integrated for 200 model years, the SST reaches a dynamic equilibrium state, with quite small fluctuation around the average value of 17.73°C , which is close to the observed value of 17.88°C in 1850 from the HadISST) [Rayner *et al.*, 2006]. FIO-ESM is quite stable and does not show much climate drift. We select model year 701 as the initial state for the historical integration of 1850–2005, and carry out the future projection runs of 2006–2100 under the four scenarios of RCP2.6, RCP4.5, RCP6.0, and RCP8.5 [Van Vuuren *et al.*, 2011].

[10] For the coupled carbon cycle model, we first integrate the ocean carbon cycle model and land carbon cycle model, separately and off-line. The ocean carbon cycle model is integrated for 3100 model years; its last 100 year averaged air-sea CO_2 flux reaches the criteria suggested by Najjar and Orr [1998], i.e., less than 0.01 PgC/yr . The land carbon cycle model is integrated for 2600 model years; its last 100 year averaged air-land CO_2 flux reaches the criteria

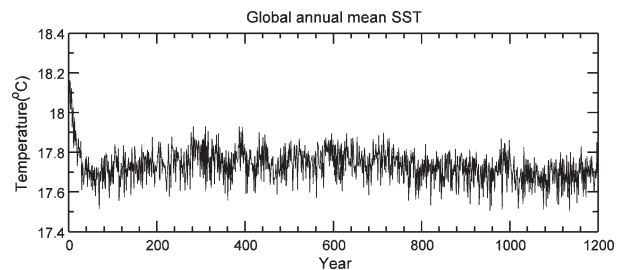


Figure 2. The evolution of the annual-mean global averaged SST in the preindustrial period from the climate model of FIO-ESM.

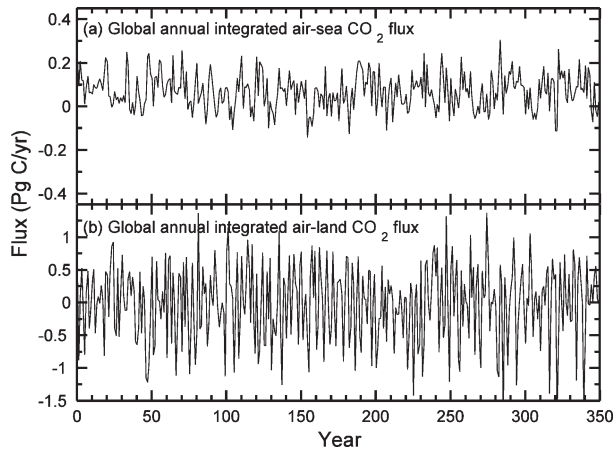


Figure 3. The annual-mean global averaged CO_2 fluxes from FIO-ESM. (top) The air-sea CO_2 flux and (bottom) the air-land CO_2 flux in the spin-up period.

suggested by Hoffman *et al.* [2008], i.e., less than 0.05 PgC/yr. Thus, the separate equilibrium states for ocean and land carbons are reached, and used as the initial conditions for the global carbon cycle model that includes ocean, land, and atmosphere components. Second, the spin-up integration for the global carbon cycle model is 350 model years, with the constant CO_2 concentration of 284.7 ppm as the forcing in the atmosphere, which is the suggested value for year 1850. The global averaged CO_2 fluxes of air-sea and air-land (Figure 3) are 0.058 ± 0.087 and 0.004 ± 0.604 PgC/yr, respectively, for the last 50 model years (301–350 model years). Third, we carry out the preindustrial control run of the global carbon cycle model for 400 model years, in which the CO_2 concentration in the atmosphere is modulated by the air-sea and air-land CO_2 fluxes and redistributed by the AGCM through equation (1). The last 50 year averaged (351–400 model years) CO_2 concentration in the atmosphere (Figure 4) is 285.2 ± 0.37 ppm, which is quite close to the observed value of 284.7 ppm in year 1850. The corresponding global averaged air-sea and air-land CO_2 fluxes are 0.033 ± 0.110 and 0.012 ± 0.582 PgC/yr, respectively. The three carbon pools of atmosphere, ocean and land reach a dynamic balance before the industrial revolution. Then, the historical run for the period of 1850–2005 and the future run for the period of 2006–2100 under the RCP8.5 scenario are conducted the same as the physical climate model, but with atmospheric CO_2 concentrations determined by the anthropogenic CO_2 emission.

3. Model Results and Validations

[11] We separate this section into three subsections. The first two subsections are physical climate model results and carbon cycle model results. Each subsection contains historical simulation of 1850–2005 and future projection of 2006–2100. In the third subsections, the wave-induced mixing effect in climate system is briefly analyzed.

3.1. Physical Climate Model Results

[12] The model results listed in this subsection are from coupled physical climate model of FIO-ESM, i.e., the carbon cycle model is not involved.

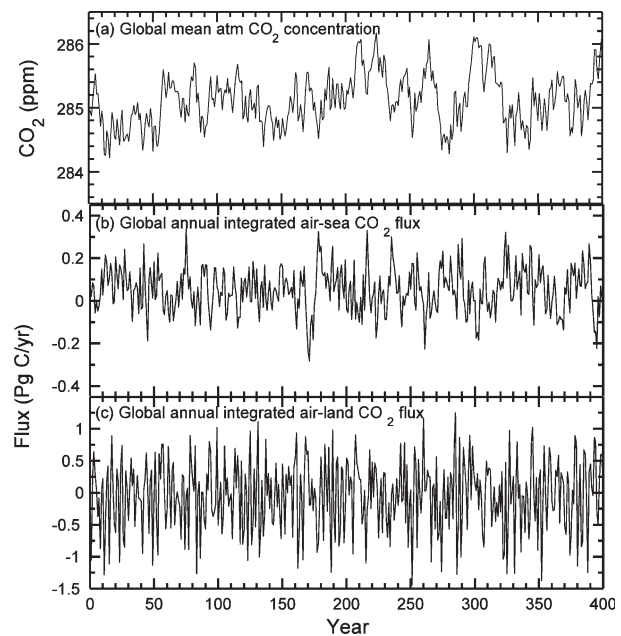


Figure 4. The FIO-ESM simulated the annual-mean global averaged CO_2 concentration in the (top) atmosphere, (middle) air-sea CO_2 flux, and (bottom) air-land CO_2 flux in the preindustrial period.

3.1.1. Historical Simulation

[13] The FIO-ESM can simulate the surface air temperature (SAT) anomaly (with respect to 1961–1990) during 1850–2005 fairly well, especially, in terms of the rapid increase after 1970 (Figure 5). During 1850 and 1950, the SAT anomaly shows an increasing tendency, though very slowly. The SAT anomaly correlation coefficient between FIO-ESM and the Met office Hadley Centre and University of East Anglia Climatic Research Unit land surface temperature dataset (CRUTEM3) [Brohan *et al.*, 2006] is 0.81 for the period of 1850–2005. However, FIO-ESM does not simulate the observed increase of SAT during 1920 and 1940 well, which is similar, if not better, to all other climate models in the CMIP5 [Jones *et al.*, 2013]. The spatial patterns of the multiyear averaged SAT during 1948–2005 from FIO-ESM simulation and National Centers for Environmental

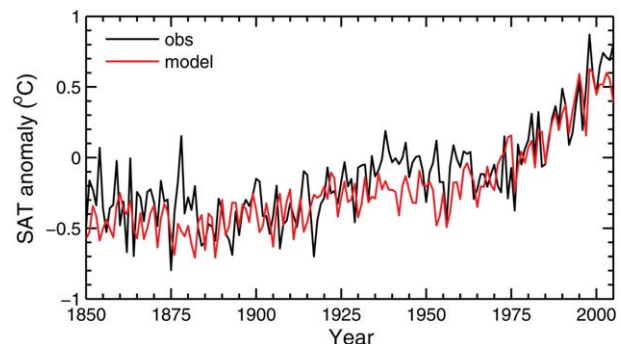


Figure 5. The evolution of the annual-mean global averaged surface air temperature anomaly during 1850–2005 from the observation of CRUTEM3 (black) and the climate model of FIO-ESM (red). The anomaly is with respect to the period of 1961–1990.

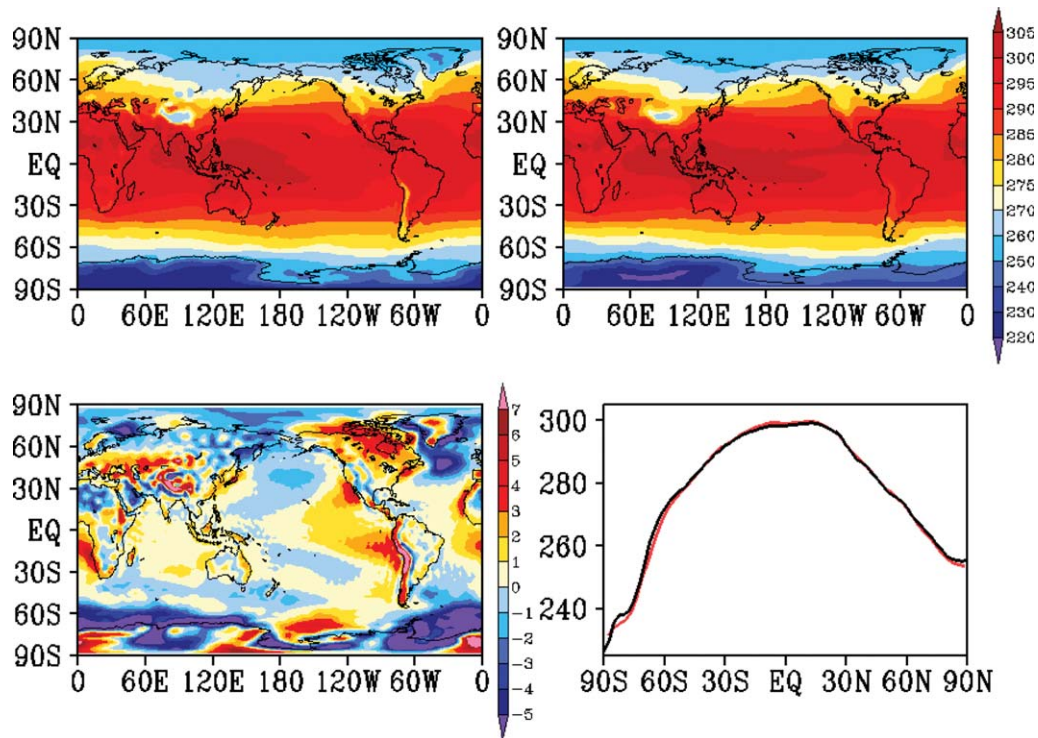


Figure 6. The multiyear averaged surface air temperature during 1948–2005 from (top left) the NCEP reanalysis, (top right) the FIO-ESM, (bottom left) the difference between FIO-ESM and NCEP, and (bottom right) zonal averaged surface air temperature (red: FIO-ESM, black: NCEP).

Prediction (NCEP) reanalysis are quite similar (Figure 6), although the simulated SAT is higher in North America and eastern parts of the Indian, Pacific, and Atlantic oceans.

[14] Another key factor for the validation of a climate model is rainfall. Since rainfall satellite data are available from 1979, the comparison between FIO-ESM output and National Oceanic and Atmospheric Administration (NOAA) Climate Prediction Center (CPC) Merged Analysis of Precipitation (CMAP) data (Figure 7) is for the period of 1979–2005. The model reproduces the observed patterns of precipitation, especially in the tropical regions, such as heavy rainfall around the western Pacific-eastern Indian Ocean warm pool and the intertropical convergence zone (ITCZ) with a narrow belt of maximum precipitation located north of equator. In the South Pacific, the simulation shows an unrealistic South Pacific convergence zone (SPCZ) parallel to the equator, but the distinctive asymmetric distribution of zonal mean precipitation over the Northern and Southern Hemispheres, which is characterized by a peak value of zonal mean rainfall in the low latitude of the Northern Hemisphere, is markedly represented in the FIO-ESM simulation. It is more comparable to the CMAP pattern, in which the peak value of zonal-mean precipitation north of the equator is greater than that in the south.

[15] The El Niño-Southern Oscillation (ENSO) is the strongest interannual variability signal in the climate system. Figure 8 shows the historical (1870–2005) ENSO wavelet analysis. The physical climate model of FIO-ESM successfully simulates a broad spectral peak in the range of 2–7 years, with double peaks around 3 and 5 years, which is basically consistent with the HadISST, although the model simulates a stronger strength for the ENSO than the

observation, i.e., the variations of amplitude in the simulation and HadISST are approximately up to 3 K and below 1 K, respectively. Numerical experiments suggest that surface waves even increase the ENSO amplitude (not shown). On the other hand, the characteristics of phase lock are well simulated, i.e., the largest SST anomalies with ENSO events occur in boreal winter, although there is spurious amplitude peak in boreal summer. In summary, FIO-ESM can simulate the basic ENSO characteristics fairly well, but the simulated strength is too strong.

3.1.2. Future Projection

[16] For CMIP5 a different experimental design was proposed in which the core simulations use prescribed Representative Concentration Pathways (RCPs) of atmospheric CO_2 and other greenhouse gases [Moss *et al.*, 2010]. The responses of global averaged SAT to the scenarios of RCP2.6, RCP4.5, RCP6.0, and RCP8.5 are shown in Figure 9. The increasing rate and variation of SAT are different under the four future scenarios. For RCP8.5, the global averaged SAT would increase by 3.92°C in 2100 (with respect to 2005) due to the greenhouse gas increase (Figure 9, left). It is interesting that the SAT would decrease after 2040 for the scenario of RCP2.6, and the global averaged SAT in 2100 would go back to the similar value as that in 2005, actually 0.007°C lower than that in 2005. For the scenarios of RCP4.5 and RCP 6.0, the global averaged SAT would increase very slowly after 2050, and become higher than the SAT in 2005 by 1.10°C and 1.85°C , respectively.

3.2. Carbon Cycle Model Results

[17] The model results listed in this subsection are from the FIO-ESM, when the carbon cycle model is

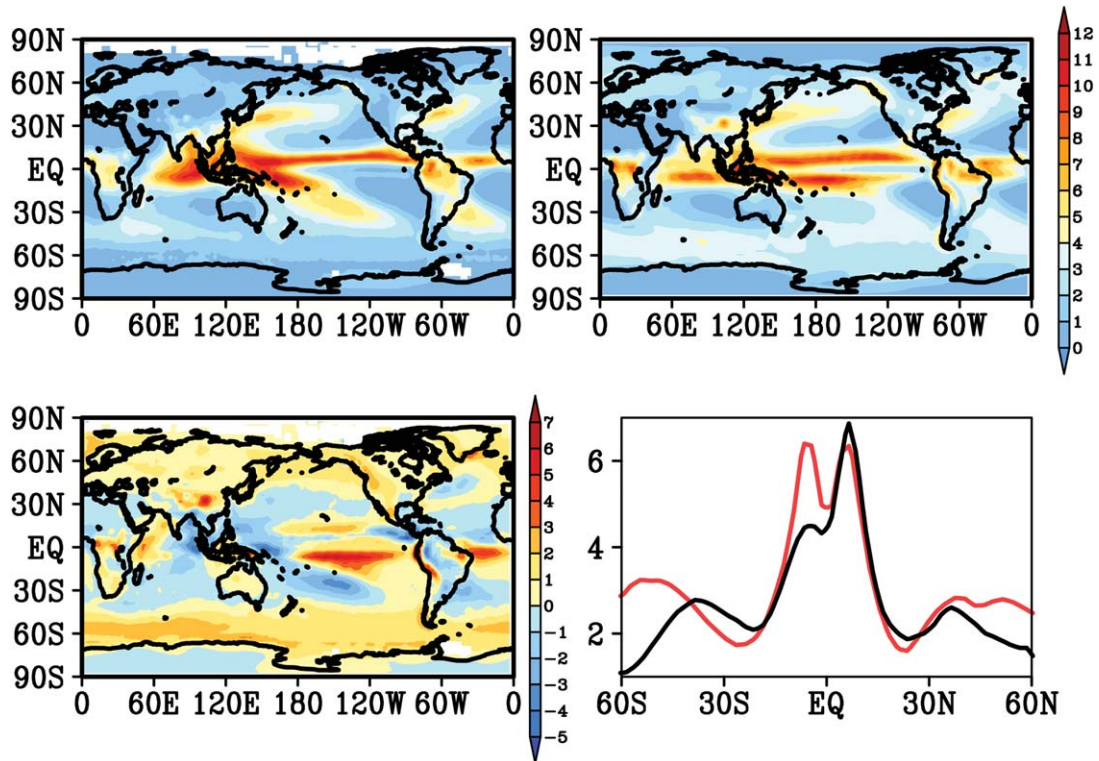


Figure 7. The multiyear averaged rainfall during 1979–2005 from (top left) the CMAP data, (top right) the FIO-ESM, (bottom left) the difference between FIO-ESM and CMAP, and (bottom right) zonal averaged precipitation rates (red: FIO-ESM, black: CMAP).

turned on and the CO_2 concentration in the atmosphere is calculated not only by the air-sea CO_2 flux and air-land CO_2 flux as in the preindustrial run but

also the anthropogenic CO_2 emissions including the fossil fuel burning emission and land use change emission.

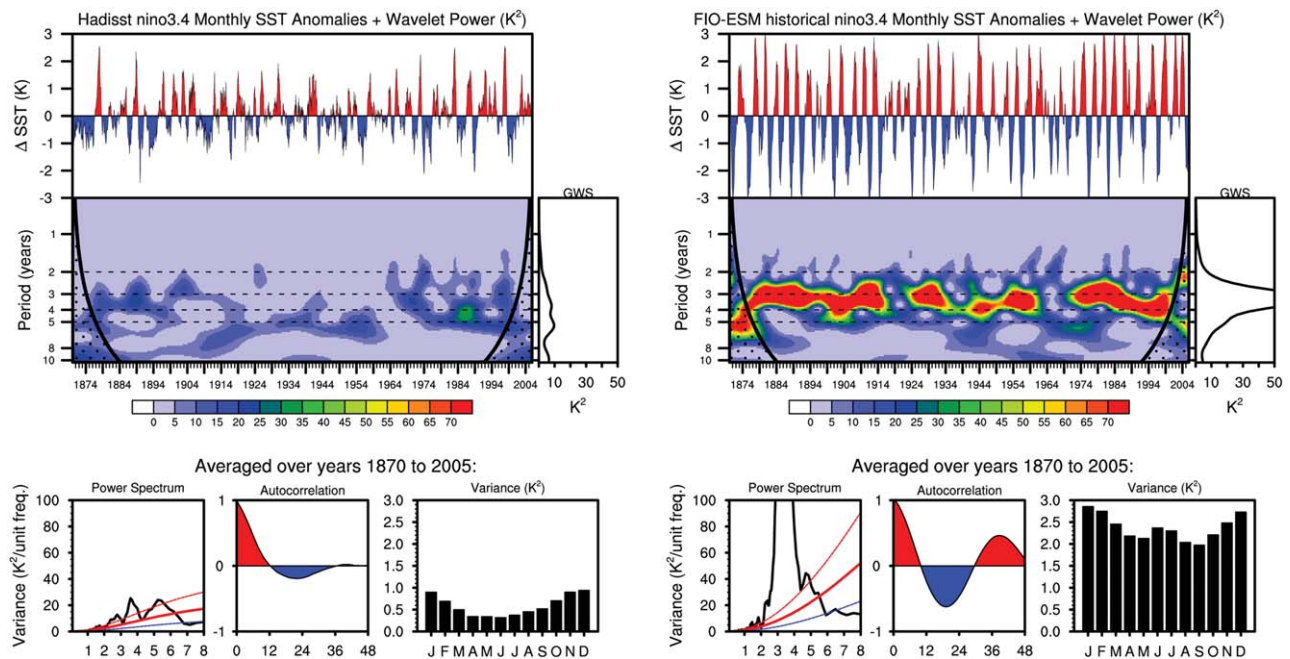


Figure 8. Wavelet analysis of historical (1870–2005) Nino3.4 SST from the (left column) HadiSST and (right column) FIO-ESM.

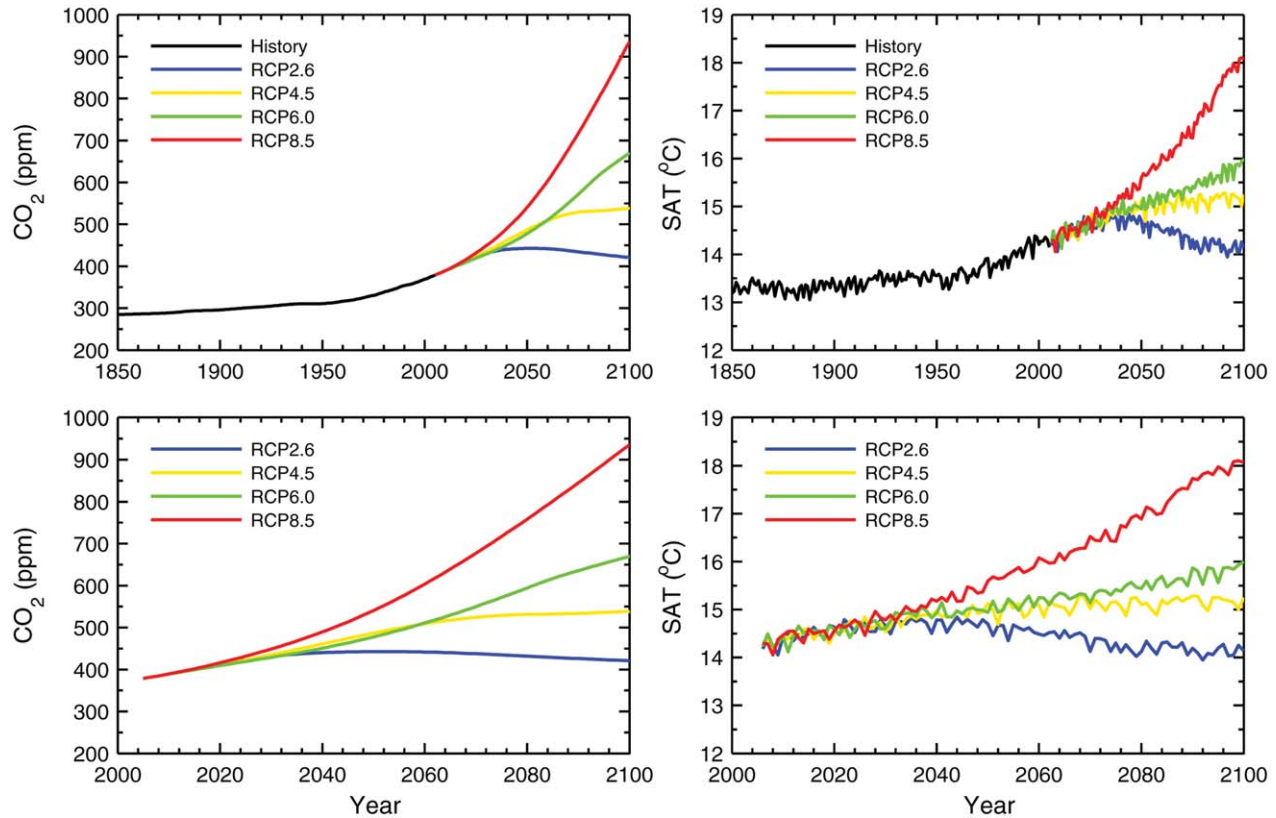


Figure 9. The evolution of the annual-mean global averaged (left) atmosphere CO_2 concentration and (right) surface air temperature from the climate model of FIO-ESM for the historical period of 1850–2005 (black) and the projections for RCP2.6 (blue), RCP4.5 (yellow), RCP6.0 (green), and RCP8.5 (red). The bottom figure zoom in to show the future projections more clearly.

3.2.1. Historical Simulation

[18] After the industrial revolution, the total anthropogenic CO_2 emission (Figure 11) increased rapidly to 8.81 PgC/yr in 2005, especially the fossil fuel burning emission reaching to 7.62 PgC/yr in 2005. The atmospheric CO_2 concentration increases as the anthropogenic emissions increase. Generally speaking, the increasing trend of the atmospheric CO_2 concentration (Figure 10) is well repro-

duced by the FIO-ESM. The simulated CO_2 concentration in the atmosphere is quite close to the observation before 1935; however, the concentration is somewhat larger than the observation after that. For a short period during 1935 and 1950, due to the net CO_2 flux into the atmosphere in the model (Figure 11), the simulated CO_2 concentration keeps on increasing while the observed remains stable. The

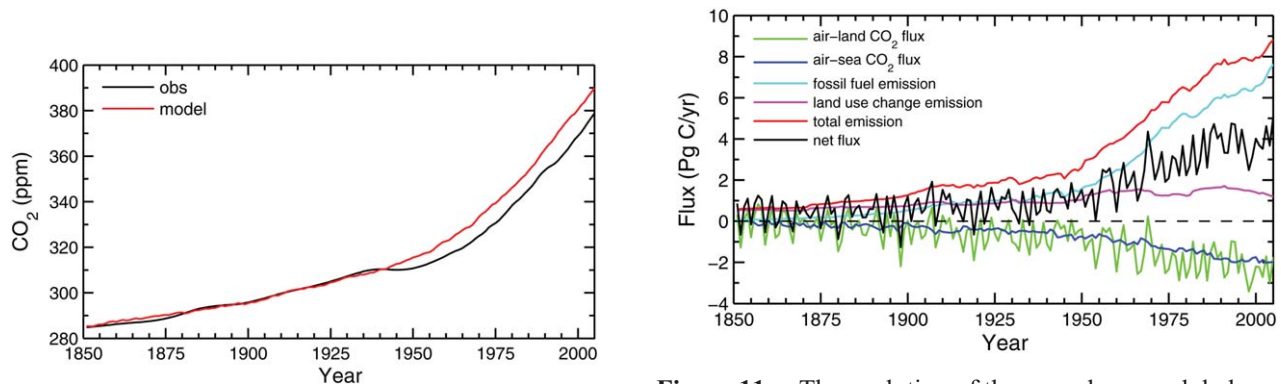


Figure 10. The evolution of the annual-mean global averaged CO_2 concentration in the atmosphere during 1850–2005 from the observation (black) and the carbon cycle model simulation of the FIO-ESM (red).

Figure 11. The evolution of the annual-mean global averaged CO_2 flux during 1850–2005 from the carbon cycle simulation of FIO-ESM, where total emission = fossil fuel emission + land use change emission; net flux = air-land flux + air-sea flux + total emission.

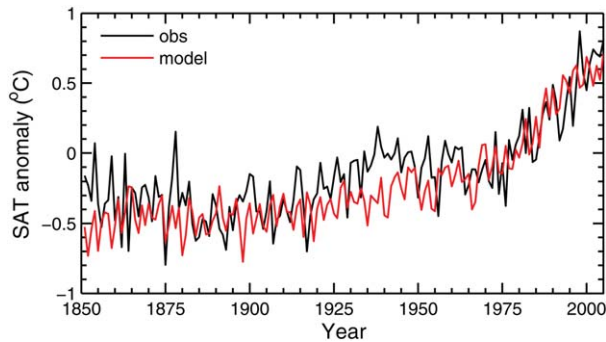


Figure 12. The evolution of the annual-mean global averaged surface air temperature anomaly during 1850–2005 from the observation of CRUTEM3 (black) and the carbon cycle model simulation of the FIO-ESM (red). The anomaly is with respect to the period of 1961–1990.

air-sea CO_2 flux is negative all the time, while the air-land CO_2 flux is negative most of the time, and positive sometimes due to its large fluctuation. This means the ocean and land absorb CO_2 from the atmosphere, and they play important roles as sinks of the anthropogenic CO_2 emissions. The averaged CO_2 sinks of ocean and land from 2000 to 2005 are 1.98 and 2.27 PgC/yr, respectively, while the averaged emission is 8.37 PgC/yr. During the historical time, the ocean and land absorb 52.5% of the anthropogenic CO_2 emissions from the atmosphere in FIO-ESM.

[19] Different from Figure 5, Figure 12 shows the SAT of FIO-ESM with the carbon cycle, i.e., the CO_2 concentration in the atmosphere for Figure 12 is from the carbon cycle model while that for Figure 5 is by forcing from observed CO_2 concentration data. The patterns in both Figures 5 and 12 are quite similar. The global averaged SAT increase, especially the sharp increase after 1970, is well simulated. However, the low temperature around 1880 and high temperature around 1940 do not appear in the model, which is the same weakness shared by other climate models and ESMs [Jones *et al.*, 2013]. The simulation of multidecadal oscillation in terms of both SAT and SST remains a challenge for all climate models.

3.2.2. Future Projection

[20] The atmospheric CO_2 concentration for RCP8.5, which is used to force the physical climate simulation, is from the prescribed concentration data of RCP8.5 scenario, while the model simulation is from the integration of FIO-ESM with the carbon cycle forced by the anthropogenic CO_2 emissions from the RCP8.5 scenario (Figure 13). Both lines are close to each other, which indicates the carbon model in FIO-ESM is reasonable. The predicted global averaged SAT would increase by 3.90°C at the end of the 21st century compared to that in 2005, the same as the result from the physical climate simulation without the global carbon cycle model.

3.3. The Effects of Nonbreaking Surface Wave in Climate Models

[21] The effects of nonbreaking surface wave-induced vertical mixing in climate system was first investigated by Song *et al.* [2007], and the common problem of the too cold tongue in climate model was much improved, the

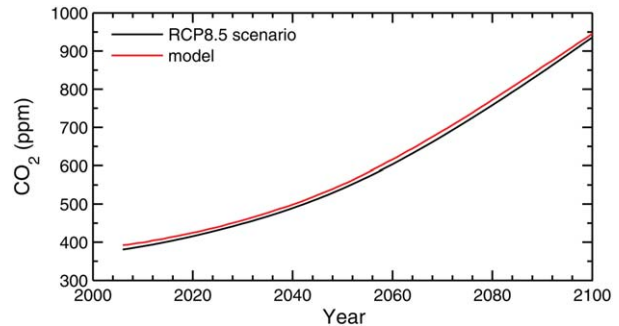


Figure 13. The evolution of the annual-mean global averaged CO_2 concentration in the atmosphere for 2006–2100 from the prescribed concentration data of RCP8.5 (black) and the carbon cycle model simulation of the FIO-ESM under RCP8.5 emissions (red).

mechanism for “West-Positive and East-Negative” SST pattern generated by Bv is investigated [Song *et al.*, 2012b], and this SST pattern is highly needed by all climate models. Babanin *et al.* [2009] tested surface wave-induced mixing parameterized in terms of the global winds in an intermediate complexity climate model, as a result the seasonal temperature modulations and extremes are significantly enhanced.

[22] The ocean MLD is one of the most important variables in the global climate system because it directly affects the air-sea fluxes of heat, freshwater, carbons dioxide, and many other properties. However, it remains a common problem that the simulated MLD in southern ocean is too shallow in austral summer. In order to investigate the wave effects in FIO-ESM, here we check the MLD simulation in physical climate model component of FIO-ESM numerical experiments with and without Bv. The MLD is defined as the depth where the seawater density is 0.03 kg/m^3 larger than that at 10 m depth. The 20 years averaged (between 1986 and 2005) model MLD of FIO-ESM without Bv in austral summer is generally shallower than observation between 50°S and 70°S (Figure 14c). It is reasonable that Bv can deepen the MLD in that zonal band. In austral winter, FIO-ESM without Bv simulates MLD deeper than observation in most areas except small area around 60°S (Figure 14a), the inclusion of Bv makes MLD shallower in most areas (Figure 14b). The mechanism needs more investigation in the future, and it should be due to some kind of air-sea feedback in climate system. Since MLD is crucial important for climate system, we gain some confidence that Bv can improve the climate model with a more realistic MLD simulation [Song *et al.*, 2012a].

[23] What’s the effect of surface wave on future climate system? We project the climate system for the period 2006–2100 based on FIO-ESM with and without Bv, respectively. The surface wave effect on the projected 20 years (2081–2100) averaged surface air temperature is defined as the difference between FIO-ESM numerical results between with and without Bv for RCP2.6 (Figure 15). Generally speaking, Bv makes surface air temperature cooler in north hemisphere, while makes it warmer in south hemisphere. In the Arctic area, Bv can make the 20 years averaged surface air temperature 2°C cooler. Different

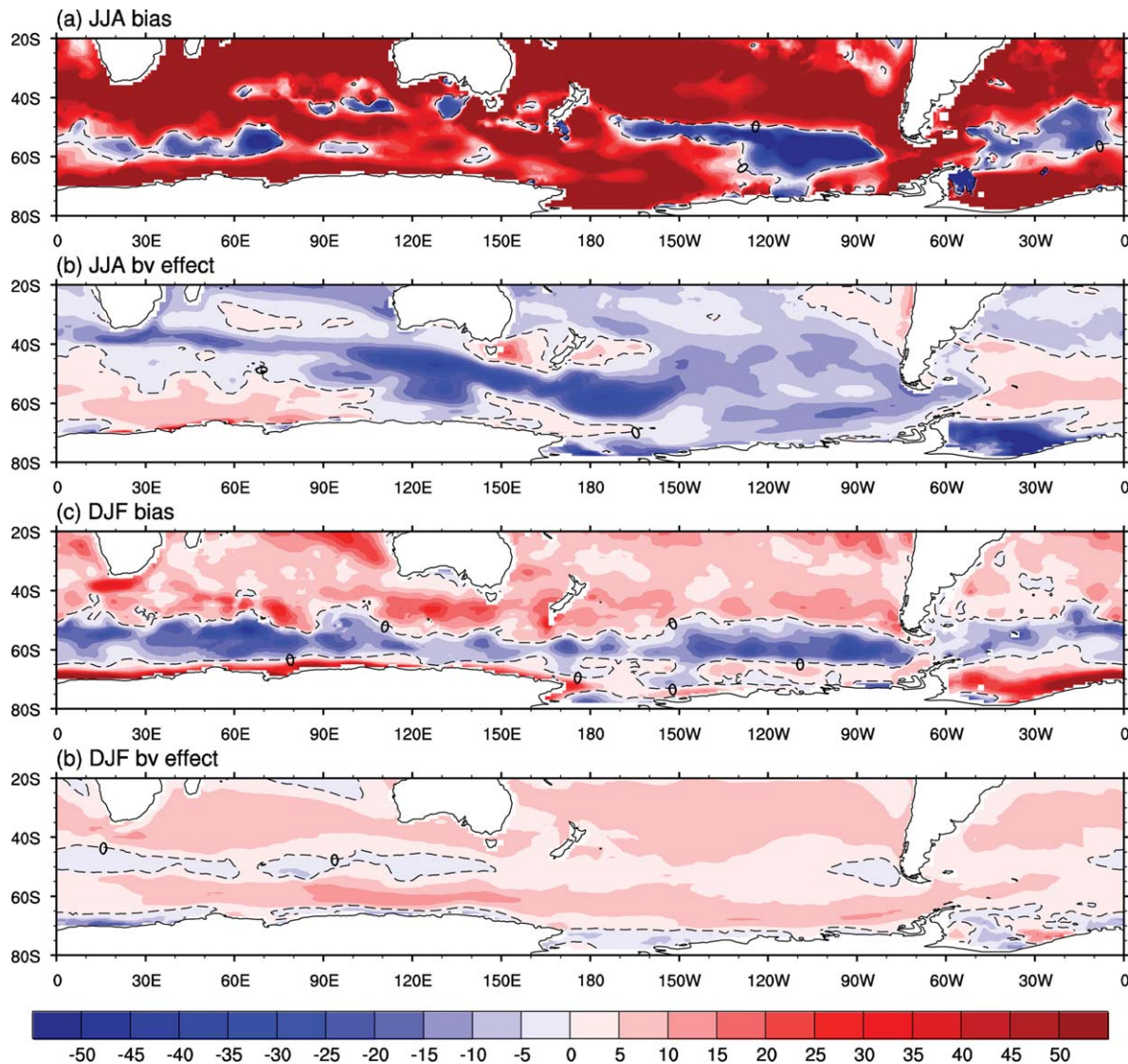


Figure 14. The simulation difference of the 20 years (1986–2005) averaged mixed layer depth in upper ocean between FIO-ESM without Bv and WOA09 in austral (a) winter and (c) summer, respectively. The surface wave effect defined as the difference between FIO-ESM numerical simulations between with and without Bv in austral (b) winter and (d) summer, respectively. Bv can improve the MLD simulation both in winter and summer.

from all other CMIP5 climate models, the sea ice in Arctic projected from FIO-ESM (with Bv) will increase for RCP2.6, RCP4.5, and RCP6.0, and decrease only for RCP8.5 (not shown).

4. Conclusions

[24] In this study, an ESM, the FIO-ESM, is established to include a physical climate model component and a carbon cycle model component. It is the first attempt to incorporate surface waves into an ESM through introducing nonbreaking wave-induced vertical mixing into an OCGM, which can improve the simulation of MLD in the southern ocean. The FIO-ESM is then employed to conduct CMIP5 numerical experiments, including the preindustrial simulation, a historical simulation of 1850–2005, and future projections for the

period of 2006–2100 under four scenarios of RCP2.6, RCP4.5, RCP6.0, and RCP8.5. The preindustrial integration was run for 1200 model years using the coupled physical climate model alone. For the FIO-ESM with the coupled carbon cycle model, we simulated 3100 and 2600 model years of ocean and land carbon cycle models off-line, respectively, then 350 model years spin-up integration of FIO-ESM with constant atmospheric CO_2 concentration forcing, followed by 400 model years of the preindustrial integration with fully coupled FIO-ESM. The four scenarios are run in the coupled physical climate simulations, while only the RCP8.5 scenario is run in the global carbon cycle simulation.

[25] Numerical experiments of coupled physical climate model of FIO-ESM show that the basic patterns of SAT, rainfall and ENSO are well simulated in the historical period, although there are still some problems, which are

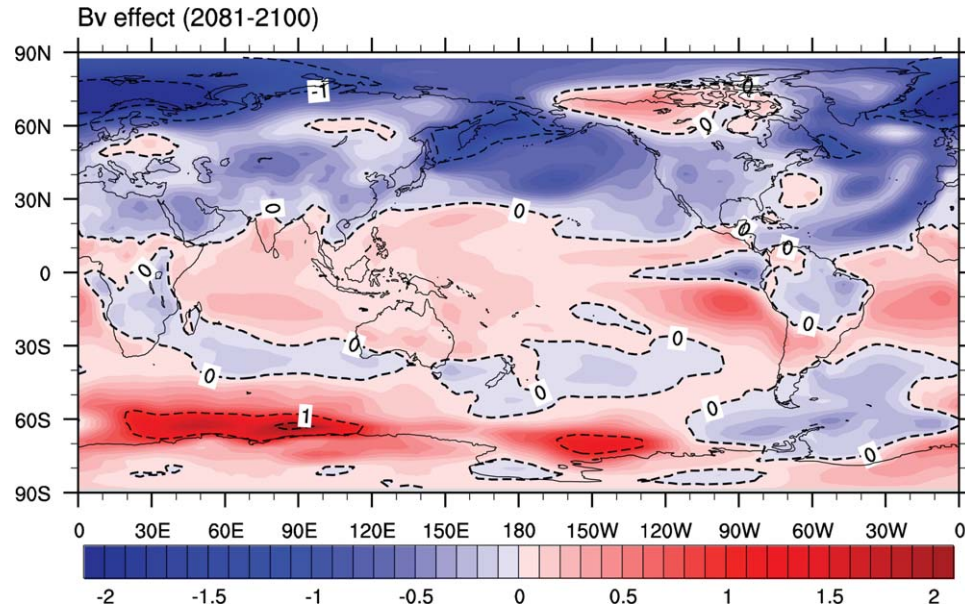


Figure 15. The surface wave effect on the projected 20 years (2081–2100) averaged surface air temperature defined as the difference between FIO-ESM numerical results between with and without Bv for RCP2.6.

common for all coupled climate models. The global averaged SAT would increase by 3.92°C in 2100 compared with that in 2005 under RCP8.5, while it would decrease 0.007°C under RCP2.6. For the global carbon simulation, the SAT and atmospheric CO_2 concentration are also well reproduced in the historical period. Model results indicate that the ocean and land play important roles as the sinks of anthropogenic CO_2 emissions. The global averaged SAT would increase by 3.90°C at the end of the 21st century (with respect to 2005) under RCP8.5, which is quite similar to the result from the global physical climate projection.

[26] Nonbreaking surface wave-induced vertical mixing makes the projected final 20 years averaged surface air temperature decrease about $0\text{--}2^{\circ}\text{C}$ in north hemisphere, and increase $0\text{--}2^{\circ}\text{C}$ in south hemisphere.

[27] All the above mentioned results suggest that the ESM of FIO-ESM incorporating surface gravity waves can reproduce the climate and global carbon cycle in the past, conduct climate future predictions, and be employed for further climate change attribution study, although there are still some common problems that are faced by all climate models. As the first step, only the surface wave-induced vertical mixing is included in the OGCM. The effects of surface waves in air-sea interaction and wave breaking-induced vertical mixing in the ocean are also important for climate system, and are considered in our ongoing research as the second step.

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